

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460169.8977 +/- 0.0078 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.064
Transit depth (Rp/Rs)^2	4.68 +/- 2.76 [%]
Orbital Inclination (inc)	88.84 +/- 2.83
Airmass coefficient 1 (a1)	1.26 +/- 0.18
Airmass coefficient 2 (a2)	-0.21 +/- 0.39
Scatter in the residuals of the lightcurve fit is	13.57 %
Best Comparison Star	#3 - [622, 52]
Optimal Aperture	6.41
Optimal Annulus	15.32
Transit Duration (day)	0.087 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.064 vs 0.1646 (deviation 0.67σ)
Duration fit vs published	0.087 d vs 0.0758 d (deviation 0.94σ)
Mid-time O–C	19.4 min
Depth SNR	2.8
Residual scatter	13.57 %

Light curve

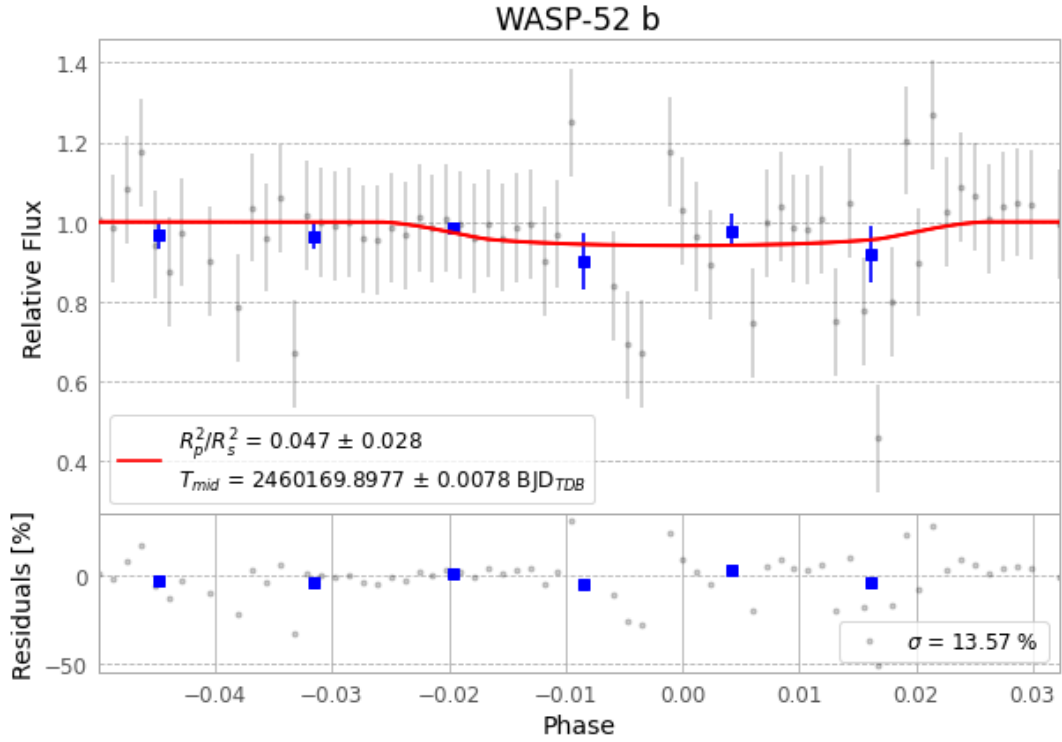


Figure 1 — FinalLightCurve_WASP-52 b_2023-08-13.png (SHA-256 dc0f1d0076834d2e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [421, 133], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 83c3aa2e4c436a1a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

69 FITS frames (45 MB), WASP-52230813071816.FITS → WASP-52230813104519.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-230813.log (fe8bf0370e04...), FinalLightCurve_WASP-52 b_2023-08-13.png (dc0f1d007683...), FinalLightCurve_WASP-52 b_2023-08-13.csv (239003ed3c83...)

Compute resources

Wall time 190.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 21:24Z.